

CS401 - Advanced Software Engineering

Individual Software Engineering Project

Module code	CS401	Module weighting	35%
Deadline	Friday 9 October 2026, 17:00	Assessment type	Individual coursework

Overview

Design and implement a production-quality web application that solves a practical problem for university students. The submission must demonstrate a structured software engineering process from requirements through evaluation.

Learning outcomes assessed

- Translate an open-ended problem into testable functional and non-functional requirements.
- Design and justify a maintainable software architecture.
- Implement a complete application using appropriate engineering practices.
- Design automated and manual tests for important behaviours and edge cases.
- Evaluate the finished system, document limitations, and propose realistic improvements.

Assessment criteria

1. Requirements analysis (15 marks)

- Identify the target user group and the specific problem being addressed.
- Define at least six functional requirements and four non-functional requirements.
- Include user stories or equivalent acceptance criteria for the core workflow.
- State assumptions, constraints and out-of-scope behaviour.

2. Architecture and system design (20 marks)

- Describe the overall architecture and responsibility of each major component.
- Include an architecture/component diagram and at least one data-flow description.
- Explain storage, service/API and client-side boundaries where relevant.
- Justify at least three significant technical decisions and discuss trade-offs.

3. Implementation (30 marks)

- Implement the core user journeys to a demonstrable standard.
- Use clear project structure, reusable components and maintainable naming.
- Validate user input and handle important failure states.
- Protect secrets/configuration appropriately and avoid committing credentials.
- Include meaningful source-control history that reflects iterative development.

4. Testing and quality assurance (20 marks)

- Provide automated tests for important deterministic or business logic.
- Perform manual testing of the primary end-to-end user journey.
- Document at least three significant bugs or regressions discovered during development.
- Show evidence that fixes were re-tested and did not break adjacent features.

5. Documentation and evaluation (15 marks)

- Provide setup and usage instructions suitable for another developer.
- Explain the final architecture and deployment approach.
- Evaluate the solution against the original requirements.
- Discuss limitations, reliability concerns and realistic future improvements.

Submission requirements

- Public or assessor-accessible source repository.
- README with setup, architecture summary and usage instructions.
- Technical report, maximum 2,500 words.
- Architecture diagram.
- Demonstration video, maximum 5 minutes.
- Live application URL where applicable.

Quality expectations

- Your submission should be internally consistent: requirements, implementation, tests and evaluation should refer to the same core behaviours.
- Where a feature is incomplete, describe the limitation accurately rather than presenting it as complete.
- Screenshots and demonstrations should show the actual submitted system.
- Evidence should be concise and relevant. Large volumes of generated documentation are not a substitute for working software or sound analysis.

Additional guidance

- Third-party APIs and AI services may be used, but their role must be clearly explained.
- Generated or external code must be acknowledged where required by module policy.
- Marks reward evidence of engineering judgement, not the number of frameworks used.

Suggested evidence checklist

- Clear repository structure and meaningful commit history.
- Working primary user journey.
- Representative success and failure cases.
- Relevant automated tests plus manual end-to-end verification.
- Architecture/deployment explanation.
- Known limitations and future improvements.

Important: this document is fictional demo material created solely for testing PlanAround. It is not a real module brief and is not issued by any university.